

ERVA SENGUL

Computer Engineer · Infrastructure & AI

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Computer engineer drawn to DevOps and AI who also enjoys building ML systems and the cloud infrastructure. Having studied in the United States at a young age, I am a native English speaker who is comfortable in international environments. Looking for a role where I can grow as an engineer across infrastructure, automation, and machine learning.

EDUCATION

Istanbul Medipol University

Istanbul, Turkiye

Bachelor's Degree, Computer Engineering GPA: 3.1/4.0

Oct 2022 - Jun 2026

- Completed a graduation project, Intelligent Cluster Optimizer a Go based Kubernetes optimizer combining a statistical engine, ML based resource forecasting, and a four-layer safety framework; published as a peer-reviewable research paper.

Relevant Coursework: DevOps, Introduction to Deep Learning, Data Communication and Computer Networks, Algorithm Analysis.

Istanbul Köy Hizmetleri High School

Istanbul, Turkiye

High School Diploma, GPA: 93.7/100

Oct 2018 - Jun 2022

PUBLICATIONS

A. Karakaya, E. Sengul, and A. Kaplan. "Safety-Gated Autoscaling: A Multi-Layered Defense Architecture for Kubernetes Vertical Resource Optimization." arXiv:2607.26503, 2026. arxiv.org/abs/2607.26503

EXPERIENCE

Medipol Healthcare Group

Istanbul, Turkiye & London, UK

Software Engineer Intern

Aug 2025 - Oct 2025

- Supported IPAM modules used in international patient operations and hospital information workflows.
- Configured, troubleshot, and supported healthcare IT systems handling secure patient data with high operational reliability.
- Visited the group's London office and saw the international patient systems in use there, gaining first hand insight into cross border healthcare operations.
- Coordinated infrastructure and system support in a production healthcare environment with strict data handling requirements.

TEKNOFEST 2025

Istanbul, Turkiye

Engineer Participant

Sep 2025

- Presented a real time emotion recognition application game built with Python, TensorFlow, Keras, and OpenCV.
- Developed a CNN pipeline for seven class facial expression recognition using CK+ data augmentation and live webcam inference.

TEI - TUSAS Engine Industries, Inc.

Istanbul, Turkiye

Student Participant

Mar 2025 - May 2025

- Completed TEI's competitive professional development program focused on aerospace systems and turbine engine engineering.
- Analyzed engineering case studies, strengthening systems thinking across reliability, performance, and technical problem solving.

Conovel Information Technologies

Istanbul, Turkiye

Software Engineer Intern

Sep 2024 - Oct 2024

- Built full-stack features in ASP.NET Core MVC using C#, HTML, CSS, and SQL.
- Implemented dynamic components and database-backed functionality to improve usability and system performance.

PROJECTS

Intelligent Cluster Optimizer | Graduation Project - Published (arXiv:2607.26503)

- Built a Go-based Kubernetes optimizer that achieved 20-40% cost savings on Google Kubernetes Engine through ML-driven vertical right-sizing.
- Designed a statistical engine, Holt-Winters forecasting, and a four-layer safety framework with memory-leak gating; validated with 1118 automated tests at 80.3% coverage.

AI Pre-Evaluation System for Medical Tourism

- Fine-tuned MedGemma-4B (QLoRA) to convert multilingual clinical documents into structured JSON; built an NLLB-200 translation pipeline with overlapping-window chunking and a deterministic clinical timeline.

Real-Time Emotion Recognition (TEKNOFEST 2025)

- Built a real-time computer-vision app classifying seven facial emotions from webcam input using Python, TensorFlow, Keras, and OpenCV.

Weather Forecast Modeling Using Interpolation

- Built a forecasting workflow on NOAA data with Python preprocessing, interpolation, and time-series analysis for seasonal pattern detection.

PIC16F877A-Based Car Battery Voltage Monitoring System

- Integrated circuit design, firmware programming, and hardware debugging for an embedded automotive monitoring prototype using PICkit 3.

TECHNICAL SKILLS

Languages: Python, Go, C#, C, SQL, JavaScript, HTML/CSS

AI / ML: Claude, Codex, Gemini, Copilot, TensorFlow, Keras, PyTorch, OpenCV, scikit-learn, Hugging Face, NLP, Computer Vision

Cloud & DevOps: Kubernetes, GKE, Docker, Git, Prometheus